Improving Temperature and Emissivity Separation Algorithm for Thermal Infrared Hyperspectral Remote Sensing: Considering Ozone Emission Effects

Xia Zhang, Ruohan Chen, Chunlai Li, Guofei Shang, Zhe Li, Yongli Chen, Haoran Xiong, Suya Zhao, and Chengyu Liu, *Member, IEEE*

***Abstract*—In thermal infrared hyperspectral remote sensing, accurate retrieval of land surface temperature (LST) and land surface emissivity (LSE) is fundamental to quantitative applications. Most existing temperature and emissivity separation (TES) algorithms primarily focus on noise suppression, adaptation to specific surface types, or improvements in computational efficiency, while relatively limited attention has been paid to retrieval disturbances caused by ozone emission. To address this issue, this study proposes the ozone-weighted TES (OWTES) algorithm, whose core concept is to enhance the weighting of ozone-sensitive spectral bands within the cost function. By strengthening the error constraints in this spectral region during the inversion process, OWTES effectively mitigates the adverse impact of noise on the retrieval procedure. Experimental results demonstrate that OWTES achieves higher retrieval accuracy for both LST and LSE, with particularly significant advantages under low-temperature and high-noise conditions. Compared with several representative TES algorithms, OWTES improves LST retrieval accuracy by an average of approximately 0.75 K and enhances LSE retrieval accuracy by about 58%. In addition, the algorithm maintains the lowest error level when validated using measured emissivity data, demonstrating high stability and reliability. These results indicate that OWTES can effectively preserve surface spectral characteristics and improve TES retrieval accuracy under complex observation conditions. Overall, the proposed OWTES method provides a more stable and reliable solution for TES in thermal infrared hyperspectral remote sensing.**

***Index Terms*—Land surface emissivity (LSE), land surface temperature (LST), ozone emission, temperature and emissivity separation (TES), thermal infrared hyperspectral remote sensing.**

Received 6 December 2025; revised 6 February 2026, 22 May 2026, and 26 June 2026; accepted 1 July 2026. Date of publication 6 July 2026; date of current version 14 July 2026. This work was supported in part by Hebei Natural Science Foundation General Program under Grant D2025403038 and in part by the National Key R&D Program of China under Grant 2022YFB3903005 and Grant 2022YFB3903005-4. *(Xia Zhang and Ruohan Chen are co-first authors.) (Corresponding authors: Chunlai Li; Chengyu Liu.)*

Xia Zhang is with Hebei Key Laboratory of Optoelectronic Information and Geo-Detection Technology and Hebei International Joint Research Center for Remote Sensing of Agricultural Drought Monitoring, Hebei GEO University, Shijiazhuang 050031, China (e-mail: zhangxia396@hgu.edu.cn).

Ruohan Chen, Guofei Shang, Haoran Xiong, and Suya Zhao are with the School of Land Science and Space Planning, Hebei GEO University, Shijiazhuang 050031, China (e-mail: chenruohan0555@163.com; shangguofei@hgu.edu.cn; xionghaoran615@163.com; zhaosuya118@163.com).).

Chunlai Li and Chengyu Liu are with the Key Laboratory of Space Active Opto-Electronics Technology, Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai 200083, China, and also with Hangzhou Institute for Advanced Study, University of Chinese Academy of Sciences, Hangzhou 310024, China (e-mail: lichunlai@mail.sitp.ac.cn; liuchengyu@mail.sitp.ac.cn).

Zhe Li is with Hebei Utilization and Planning Institute of Natural Resources, Shijiazhuang 050011, China (e-mail: 13803117867@163.com).

Yongli Chen is with the Second Surveying and Mapping Institute of Hebei Province, Shijiazhuang 050031, China (e-mail: 73cyl@163.com).

Digital Object Identifier 10.1109/TGRS.2026.3710493

I. Introduction

THERMAL infrared hyperspectral remote sensing, benefiting from its exceptionally high spectral resolution, has demonstrated irreplaceable value in various fields, including resource exploration [1], [2], [3], surface environmental monitoring [4], [5], [6], [7], and atmospheric pollution detection [8], [9], [10]. For instance, thermal infrared hyperspectral data enable the identification of diagnostic spectral features of specific minerals, particularly silicate minerals such as quartz and feldspar, thereby supporting accurate mineral mapping [11], [12], [13]. In environmental monitoring, such data can be used to retrieve key biophysical parameters, such as soil moisture and vegetation growth conditions, facilitating refined vegetation monitoring [14], [15]. In atmospheric applications, thermal infrared hyperspectral observations capture the spectral signatures of gaseous pollutants, enabling both source localization and concentration estimation [16], [17]. These capabilities arise from the unique advantage of hyperspectral sensing, which provides dense and continuous spectral information, allowing detailed characterization of surface and atmospheric properties beyond what broadband or multispectral sensors can achieve.

The effectiveness of these applications fundamentally depends on the accurate retrieval of land surface temperature (LST) and land surface emissivity (LSE) from measured radiance. LST reflects the surface energy balance and thermodynamic state, while LSE is determined by the intrinsic physical and chemical properties of surface materials and therefore serves as a key parameter for material identification and classification. In particular, emissivity contains the spectral signatures that enable hyperspectral systems to distinguish different surface types. Any error in emissivity directly propagates into errors in both temperature retrieval and material interpretation, significantly degrading the quantitative capability of thermal infrared hyperspectral remote sensing.

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.

Therefore, accurate separation of temperature and emissivity is not only a prerequisite for reliable physical interpretation but also a determining factor for the overall performance of hyperspectral applications.

However, temperature–emissivity separation (TES) remains a fundamental challenge due to its inherently ill-posed nature. Specifically, the radiance measured by a hyperspectral sensor provides N spectral observations, whereas the retrieval requires solving for N emissivity values together with one surface temperature, resulting in N+ 1 unknowns. This mismatch leads to an underdetermined system in which a unique and stable solution cannot be obtained without introducing additional constraints or parameterizations. Consequently, extensive research efforts have been devoted to developing TES algorithms that incorporate physical assumptions or mathematical regularization to stabilize the inversion process.

Among the various TES approaches, several representative methods have played a central role in advancing the field. The Iterative Spectrally Smooth TES (ISSTES) algorithm [18] is a classical method that exploits the spectral smoothness of emissivity. It is based on the physical observation that emissivity spectra, governed by material composition, tend to be smooth and continuous, whereas atmospheric downwelling radiance contains numerous narrow absorption features caused by gases such as water vapor and carbon dioxide. By constructing a smoothness-based cost function that evaluates residual atmospheric absorption features in the retrieved emissivity spectrum, ISSTES determines the optimal temperature through iterative minimization and subsequently derives the emissivity.

Building on this concept, the Automatic Retrieval of Temperature and Emissivity using Spectral Smoothness (ARTEMISS) algorithm [19] improves the robustness of TES by reformulating the optimization criterion. Instead of directly relying on a smoothness index, ARTEMISS determines the optimal temperature by minimizing the discrepancy between simulated and measured radiance. This formulation reduces sensitivity to local spectral fluctuations and improves stability under complex atmospheric conditions. To further enhance computational efficiency for practical applications, ARTEMISS has also been extended to fast-search strategies such as QTES [20], although the core inversion principle remains based on spectral smoothness.

To improve stability in the presence of noise, the Resolution-Degraded Spectral Smoothing (RDSS) method [21] introduces a smoothing operator into the radiative transfer equation. By applying spectral smoothing to both radiance and emissivity-related variables, RDSS suppresses high-frequency noise and stabilizes the inversion process, leading to improved accuracy for both LST and LSE retrievals. This approach highlights the importance of noise handling and regularization in TES algorithms.

In contrast to these spectral-smoothness-based methods, the Wavelet-Transform TES (WTTES) algorithm [22] adopts a different strategy by reducing the dimensionality of the emissivity spectrum. Through wavelet decomposition, emissivity is represented using a limited number of low-frequency coefficients, thereby reducing the number of unknowns from N+ 1 to a smaller set of parameters. This dimensionality-reduction approach alleviates the ill-posedness of TES and provides a more tractable inversion framework, offering an alternative pathway for TES.

Despite these advances, existing TES methods still suffer from several inherent limitations. First, most algorithms—including ISSTES, ARTEMISS, and RDSS—are fundamentally built upon a common assumption that emissivity spectra are smooth while atmospheric absorption features are narrow and localized. This assumption underlies the design of cost functions and temperature-search strategies in these methods. While it is valid under idealized conditions, it does not fully capture the complexity of real atmospheric radiative transfer processes.

Second, although dimensionality-reduction approaches such as WTTES reduce the number of unknowns and improve numerical stability, they introduce additional modeling assumptions regarding the representation of emissivity spectra. These assumptions may not hold for complex or heterogeneous surfaces, potentially limiting the general applicability of the method.

More importantly, most existing TES studies focus on improving inversion performance through general strategies such as noise suppression, parameter simplification, or computational optimization. In contrast, relatively limited attention has been paid to how specific atmospheric absorption characteristics influence the inversion process itself. This gap becomes particularly critical when atmospheric spectral structures deviate from the assumptions embedded in current TES algorithms.

One notable example is ozone absorption in the thermal infrared region. Ozone exhibits a pronounced broadband absorption feature spanning approximately 9.2–10.0 μm, which introduces a slowly varying spectral component into the atmospheric downwelling radiance. This broadband structure is fundamentally different from the narrow, spike-like absorption features typically assumed in spectral-smoothness-based TES algorithms. As a result, ozone-induced spectral features can distort the cost-function evaluation during temperature retrieval, leading to confusion between atmospheric effects and surface emissivity characteristics. Such misinterpretation weakens the effectiveness of smoothness constraints and introduces systematic errors in both LST and LSE retrievals.

Motivated by this limitation, this study aims to explicitly investigate the impact of ozone absorption on the TES process and to improve the formulation of the cost function accordingly. By re-examining the role of spectral structures in temperature-search strategies, we propose a differentiated-weight cost function that enhances the contribution of ozone-sensitive spectral regions. Based on this idea, an ozone-weighted TES (OWTES) method is developed. The proposed method quantitatively determines the optimal spectral weighting for ozone-affected bands and aims to improve retrieval accuracy under realistic atmospheric conditions, thereby providing a more physically consistent solution to the TES problem.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.

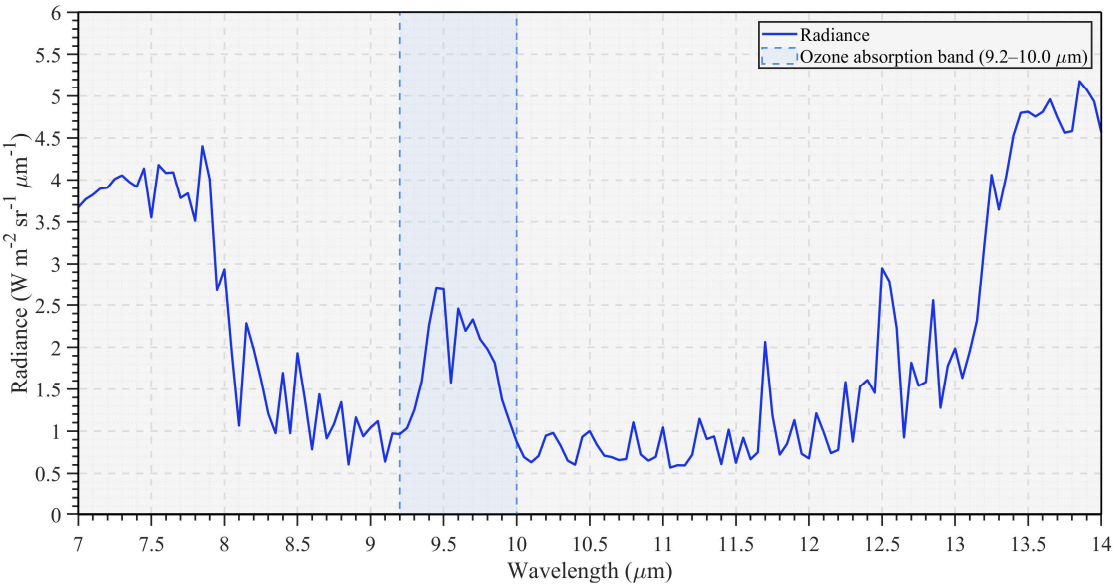


Fig. 1. Atmospheric downwelling radiance spectra under the MLS atmosphere in MODTRAN.

II. Development of the OWTES Algorithm

In thermal infrared remote sensing, the at-sensor radiance measured by sensors onboard remote sensing platforms is commonly described using a simplified form of the atmospheric radiative transfer equation under standard assumptions. The radiative transfer equation in the thermal infrared region can be expressed as

$$L(\lambda_k, T) = L_g(\lambda_k, T)\, t(\lambda_k) + L_u(\lambda_k) \tag{1}$$

$$L_g(\lambda_k, T) = \varepsilon(\lambda_k)\, B(\lambda_k, T) + [1 - \varepsilon(\lambda_k)]\, L_d(\lambda_k) \tag{2}$$

where, for the kth spectral band, $L(\lambda_k, T)$ represents the at-sensor radiance when the surface temperature is T; $t(\lambda_k)$ is the atmospheric transmittance; $L_u(\lambda_k)$ is the atmospheric upwelling radiance; $L_g(\lambda_k, T)$ denotes the ground-leaving radiance at wavelength λ_k for surface temperature T;$\varepsilon(\lambda_k)$ is the LSE;$B(\lambda_k, T)$ is the blackbody radiance at temperature T; and $L_d(\lambda_k)$ is the atmospheric downwelling radiance.

During the radiative transfer process, the spectral characteristics of atmospheric downwelling radiance varied across different wavelength regions, thereby exerting distinct influences on LST–LSE retrieval. When the noise level was low, the spike-like structures in the spectrum effectively facilitated the separation of surface-emitted radiance from atmospheric contributions. However, under high-noise conditions, noise interacted with these spectral structures and substantially reduced the separability between surface blackbody radiance and atmospheric downwelling radiance. In retrieval algorithms based on radiative transfer modeling, this reduced separability directly weakened the sensitivity of the cost function to the true LST and LSE. As a consequence, the optimization process often converged to incorrect parameter values, stagnated, or deviated from physically valid solutions, ultimately leading to significant degradation in LST accuracy and large biases in LSE estimation.

At this stage, the influence of the ozone absorption band became increasingly pronounced. As shown in Fig. 1, under the mid-latitude summer (MLS) atmospheric profile, the atmospheric downwelling radiance spectrum exhibited a distinct ozone absorption feature within the 9.2–10.0 μm range. This feature suppressed the spike-like structures typically observed in atmospheric spectra and resulted in a relatively smooth broadband signature, thereby facilitating the separation of surface-emitted radiance from atmospheric downwelling radiance. Consequently, it became necessary to quantify the contribution of ozone-sensitive wavelengths to the retrieval process by adjusting their weights within the cost function. Based on this consideration, the cost function was modified accordingly

$$\sigma_{\mathrm{OWTES}}\left(\hat{T}\right) = \sqrt{\frac{1}{k}\sum_{k=1}^{k} \omega(\lambda_k) \cdot \left[L_{g,\mathrm{sim}}\left(\lambda_k, \hat{T}\right) - L_g(\lambda_k, T)\right]^2} \tag{3}$$

$$\omega(\lambda_k) = \begin{cases} a, & \lambda_k \in (9.2\mu\mathrm{m}, 10\mu\mathrm{m}) \\ 1, & \lambda_k \notin (9.2\mu\mathrm{m}, 10\mu\mathrm{m}) \end{cases} \tag{4}$$

$$L_{g,\mathrm{sim}}\left(\lambda_k, \hat{T}\right) = \tilde{\varepsilon}(\lambda_k)\, B\left(\lambda_k, \hat{T}\right) + [1 - \tilde{\varepsilon}(\lambda_k)]\, L_d(\lambda_k) \tag{5}$$

where $\omega(\lambda_k)$ denotes the weight assigned to each spectral band and a represents the weighting coefficient assigned to the ozone-sensitive spectral region (9.2–10.0 μm). By adaptively adjusting the weights of ozone-sensitive wavelengths within the cost function, the retrieval performance can be optimized. $L_{g,\mathrm{sim}}(\lambda_k, \hat{T})$ represents the simulated ground-leaving radiance, while $\hat{T}$ and $\tilde{\varepsilon}(\lambda_k)$ denote the retrieved LST and emissivity, respectively. In the OWTES algorithm, we applied the RDSS approach [21] to reduce noise. The flowchart of OWTES is illustrated in Fig. 2, and the procedure is as follows.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.

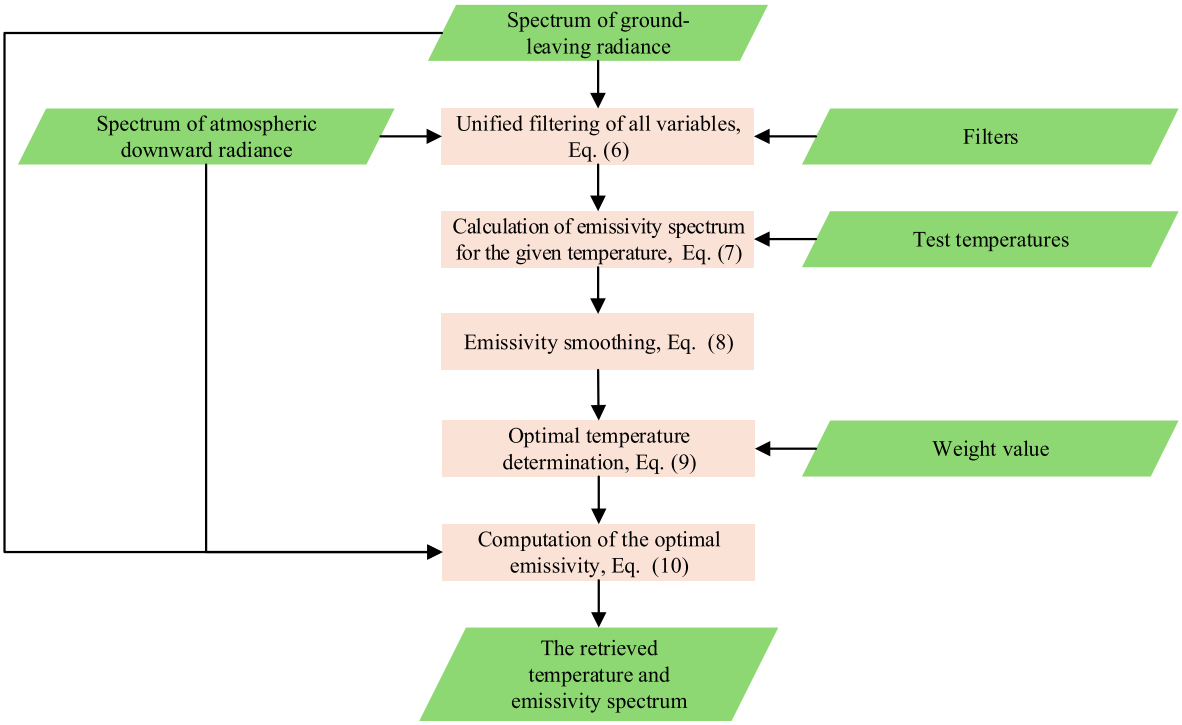


Fig. 2. Flowchart of the OWTES algorithm.

1) *Unified Filtering of all Variables*: Before computing the emissivity, a mean filter is applied to each variable in the ground-leaving radiance equation in order to suppress high-frequency noise and narrow spectral fluctuations, thereby improving the stability of the emissivity retrieval. The variables in (2) are processed using a mean filtering function, which is computed as follows:

$$\tilde{L}_g(\lambda_k, T) = \frac{1}{3}\sum_{j=-1}^{1}\left[L_g\left(\lambda_{k+j}, T\right)\right] \tag{6}$$

where $\tilde{L}_g(\lambda_k, T)$ denotes the mean-filtered ground-leaving radiance $L_g(\lambda_k, T)$. The size of the filtering window determines the effectiveness of the smoothing operation in suppressing spectral spikes. When the window is excessively large, spike features may be completely removed, resulting in distortion of the spectral curve. Conversely, if the window is too small, residual spikes may remain. Previous simulation studies [19], [21] have shown that a window radius of 3 results in the lowest retrieval error.

2) *Emissivity Computation for a Given Temperature*: Because temperature and emissivity are uniquely coupled through the radiative transfer equation, the filtered variables can be substituted into (5), from which the emissivity is computed as

$$\hat{\varepsilon}(\lambda_k) = \frac{\tilde{L}_g(\lambda_k, T) - L_d(\lambda_k)}{B\left(\lambda_k, \hat{T}\right) - L_d(\lambda_k)}. \tag{7}$$

3) *Emissivity Smoothing*: To suppress spike-like artifacts introduced by atmospheric absorption and noise, and to ensure that the retrieved emissivity closely conforms to its expected spectral behavior, a mean-filtering operation is applied to smooth the emissivity spectrum. The smoothing process is computed as follows:

$$\tilde{\varepsilon}(\lambda_k) = \frac{1}{3}\sum_{j=k-1}^{k+1}\hat{\varepsilon}(\lambda_k) \tag{8}$$

where $\tilde{\varepsilon}(\lambda_k)$ denotes the smoothed emissivity and the value 3 represents the radius of the filtering window.

4) *Optimal Temperature Determination*: The optimal temperature is determined by iteratively evaluating the cost function and selecting the temperature at which the cost function attains its minimum value (9) and (10), as shown at the bottom of the next page.

5) *Computation of the Optimal Emissivity*: The optimal temperature estimate is substituted into (10), and the final emissivity is computed using the unfiltered variables. The resulting expression is as follows:

$$\hat{\varepsilon}_{\text{opt}}(\lambda_k) = \frac{L_g(\lambda_k, T) - L_d(\lambda_k)}{B\left(\lambda_k, T_{opt}\right) - L_d(\lambda_k)} \tag{11}$$

where $\hat{\varepsilon}_{\text{opt}}(\lambda_k)$ denotes the final retrieved emissivity.

III. Data and Data Processing

A. Simulated Data

Thermal infrared hyperspectral imagers serve as the primary instruments for acquiring thermal infrared hyperspectral remote sensing data. To simulate sensor-measured radiance and comprehensively represent the disturbances and parameter variations encountered in real-world applications, the influencing factors and input parameters in this study are grouped into two categories. The first category includes three factors that are independent of the core TES algorithm and characterize

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.

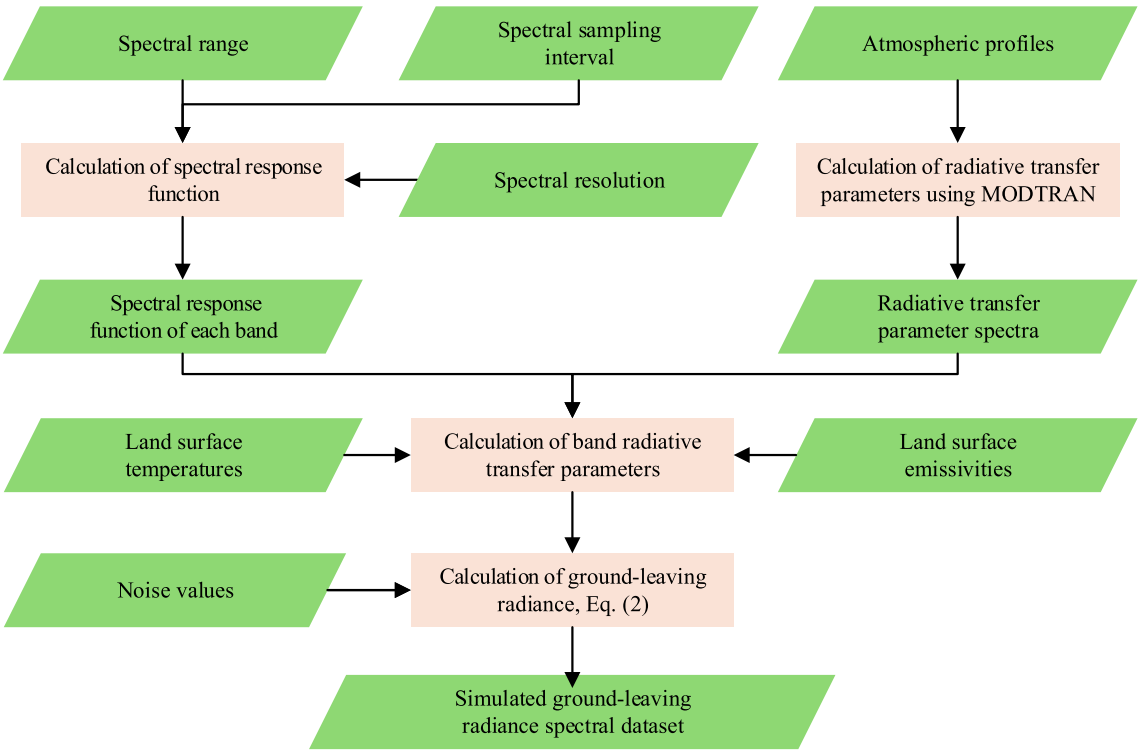


Fig. 3. Flowchart of the simulated ground-leaving radiance generation.

TABLE I
PRIMARY VARIABLES SETUP OF THE SIMULATION EXPERIMENT

Variable	Value	Number
Spectral range	8-12.5μm	1
Spectral sampling interval	25 nm	1
Random noise	From 0 to 0.1K with an incremental step of 0.05K From 0.1 to 1.0K with an incremental step of 0.1K	12
LST	Lower atmospheric temperature + offset Offset: from −15K to 30K with an incremental step of 5K	10
LSE	All emissivity spectra covering thermal infrared wavelengths from ASTER 2.0	1241
Standard Atmospheric profiles	1976 US Model, 288.15K Tropical Model, 299.7K Mid-Latitude Summer Model, 294.2K Mid-Latitude Winter Model, 272.2K Sub-Arctic Summer Model, 287.2K Sub-Arctic Winter Model, 257.2K	6
Repetition	—	100
Total	—	89,352,000

only the data acquisition stage: spectral range, spectral resolution, and noise level. The second category comprises four parameters that directly participate in radiative transfer modeling and fundamentally determine the inversion conditions:

$$T_{opt} = \arg\min_{\hat{T}} \left(\sigma_{\mathrm{OWTES}}\right) \tag{9}$$

$$\sigma_{\mathrm{OWTES}}\left(\hat{T}\right) = \sqrt{\frac{1}{N}\sum_{k=1}^{N} \omega\left(\lambda_k\right) \cdot \left[\tilde{\varepsilon}\left(\lambda_k\right) B\left(\lambda_k, \hat{T}\right) + \left[1 - \tilde{\varepsilon}\left(\lambda_k\right)\right] L_d\left(\lambda_k\right) - L_g\left(\lambda_k, T\right)\right]^2} \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.

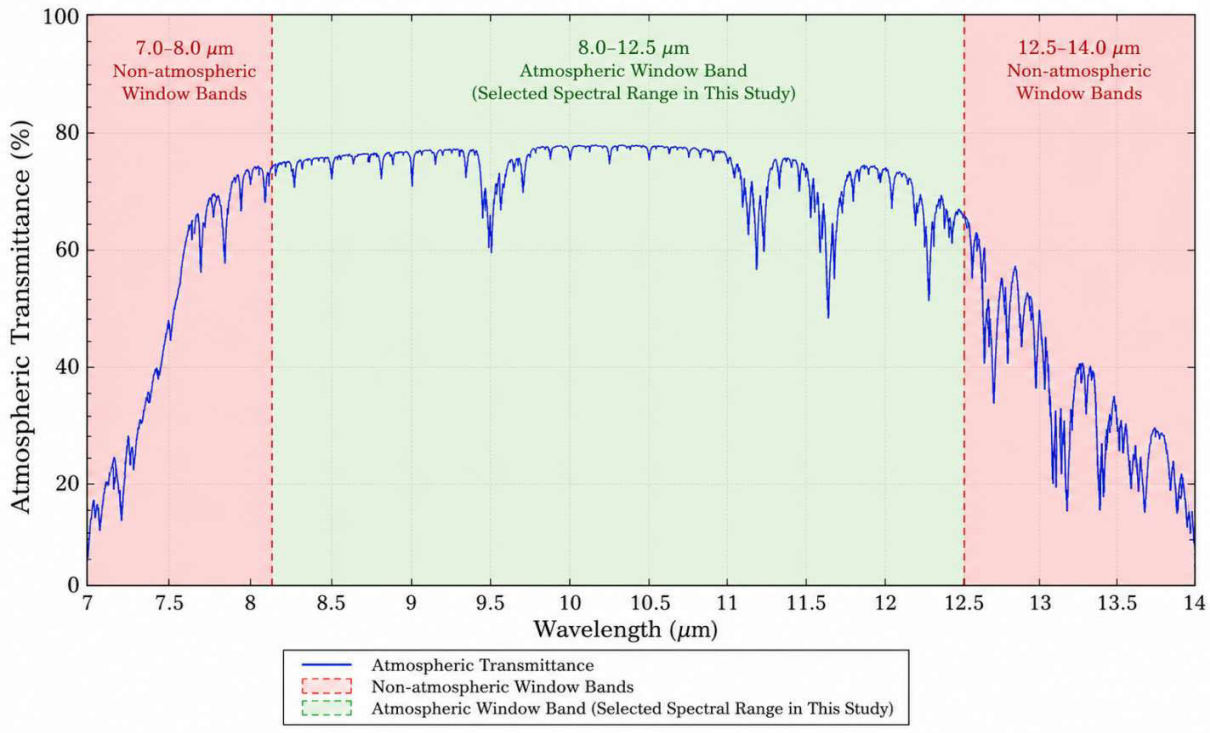


Fig. 4. Atmospheric transmittance characteristics and selected spectral range for this study.

LST, LSE, and the atmospheric model. The detailed configuration of all parameters is summarized in Table I, and the workflow of ground-leaving radiance simulation is illustrated in Fig. 3.

1) Spectral Range: In thermal infrared remote sensing, the 7.5–13.5 μm region is commonly regarded as the primary atmospheric window, where atmospheric attenuation is relatively weak, making it suitable for remote-sensing observations. However, for practical temperature retrieval using thermal infrared hyperspectral systems, the usable spectral range must be refined according to atmospheric interference characteristics. The 7.5–8.0 μm interval contains dense water-vapor absorption lines, which significantly enhance selective absorption, degrade the signal-to-noise ratio (SNR), distort radiometric information, and introduce biases into temperature-retrieval models. In extreme cases, these effects can lead to abnormal inversion results. To mitigate these issues, this study defines the experimental spectral range as 8–12.5 μm. This interval effectively avoids the strongly absorbing region while maintaining radiative stability within the core thermal infrared atmospheric window. The atmospheric transmittance characteristics and the selected spectral range are illustrated in Fig. 4.

2) Spectral Resolution: The hyperspectral thermal infrared (HTIR) imager employs spectral-splitting technology to divide the thermal infrared spectrum into hundreds to thousands of nearly continuous narrow bands. The center wavelengths were uniformly distributed within the 8.0–12.5 μm spectral range with a sampling interval of 25 nm, while the full width at half maximum (FWHM) of each band was set to 50 nm. The spectral response of each channel can be well approximated by a Gaussian distribution. In practical applications, after spectral calibration using instruments such as monochromators, the spectral response characteristics of each channel are typically modeled by fitting a Gaussian function. Accordingly, the spectral response function of each band was represented by a Gaussian curve defined by its center wavelength and FWHM. Most current airborne thermal infrared hyperspectral sensors achieve spectral resolutions of 50 nm or better. Therefore, this study adopted a spectral resolution of 50 nm for all experiments, which is representative of mainstream HTIR imagers. This setting is suitable for evaluating the influence of spectral resolution on retrieval performance and provides targeted support for optimizing TES algorithms based on existing instrument capabilities.

3) Noise Level: Calibration data acquired by HTIR imagers contain noise originating from multiple sources, including intrinsic detector noise, electronic transmission noise, and radiometric fluctuations of the calibration source. Calibration errors generally comprise systematic and random components; however, because random errors lack stable or predictable patterns and cannot be effectively eliminated through standard calibration procedures, this study focuses exclusively on their impact.

In this study, noise is quantified using the noise equivalent delta temperature (NEDT), which characterizes the random error component in calibration data and is modeled using a Gaussian distribution. To ensure that the simulated data cover realistic noise conditions, we define 12 noise levels spanning 0–1 K: 0–0.1 K in increments of 0.05 K and 0.1–1 K in increments of 0.1 K. This configuration enables

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.

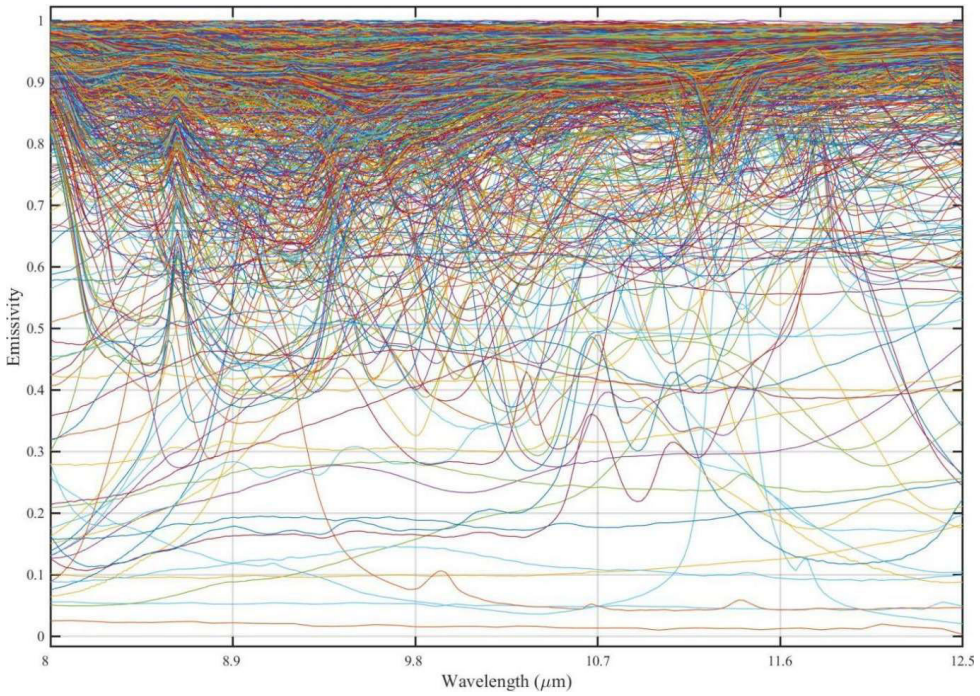


Fig. 5. 1421 emissivity curves from ASTER 2.0 Spectral Library (8–12.5 μm).

a comprehensive assessment of noise effects across both low- and high-noise regimes.

4) Land Surface Temperature: Frequent heat exchange occurs between the land surface and the atmosphere, and LST generally exhibits greater variability than the temperature of the lower atmosphere. Therefore, in simulation experiments, LST is specified relative to the corresponding atmospheric temperature using the form "atmospheric temperature + offset." In this study, the offset range is set from −15 to 30 K with an increment of 5 K.

5) Land Surface Emissivity: Spectral-smoothing-based TES algorithms are founded on the principle that LSE spectra exhibit greater smoothness than atmospheric downwelling radiance. Therefore, a sufficiently diverse set of emissivity spectra is required to evaluate their influence on retrieval accuracy. In this study, we employ the ASTER 2.0 spectral library, which is widely used in remote sensing simulation studies. The library contains 2445 land-surface spectra, among which 1241 emissivity spectra within the thermal infrared region are selected to represent typical application scenarios. These samples cover major surface types, including minerals (737), rocks (386), man-made materials (53), soils (52), vegetation (4), water (4), snow (4), and ice (1). Representative emissivity spectra of these surface materials are shown in Fig. 5.

6) Atmospheric Profiles: To represent a broad range of atmospheric conditions, six atmospheric models were selected from the MODTRAN medium-resolution atmospheric transmission software developed by Burlington Spectral Sciences, Inc. These include the 1976 U.S. Standard Atmosphere, the tropical atmosphere, the MLS and winter atmospheres, and the subarctic summer and winter atmospheres.

This study adopts the following method to determine the optimal temperature for the OWTES algorithm. The ground-leaving radiance simulation workflow is presented in Fig. 2, with detailed parameter configurations summarized as follows.

1) *Smoothing Window Size*: According to the analysis in [28], which evaluated the effect of spectral-smoothing window size in the ARTEMISS algorithm using 17 different window settings, a window size of 3 yielded the smallest retrieval error under low-noise conditions. Accordingly, the spectral-smoothing window size is set to 3 in this study.

2) *Optimal Temperature Search Strategy*: To reduce computational complexity and avoid convergence of the cost function to local minima, this study adopts the optimal temperature-search strategy described in [23]. In this strategy, the initially estimated minimum temperature is used as the lower bound of the search interval, while the upper bound is defined as 60 K above it, thereby reducing the invalid search range and improving temperature retrieval efficiency. The temperature search step size is set to 0.1 K. Within this search interval, the cost function is evaluated at each temperature, and the temperature corresponding to the minimum cost value is selected as the optimal solution.

The root-mean-square error (RMSE) is used to evaluate the retrieval accuracy of both LST and LSE. The RMSE for a set of LST and LSE values under a given spectral configuration and noise level is defined as

$$RMSE_T = \sqrt{\frac{1}{N_s}\sum_{j=1}^{N_s}\left(\hat{T}_{\text{opt}} - T_j\right)^2} \tag{12}$$

$$rmsE_\varepsilon = \sqrt{\frac{1}{N_S N}\sum_{j=1}^{N_s}\sum_{i=1}^{N}\left(\hat{\varepsilon}_{\text{opt}}(\lambda_i) - \varepsilon_j(\lambda_i)\right)^2} \tag{13}$$

where $j \in [1, N_s]$, N_s denotes the number of samples in the dataset under evaluation; $\hat{T}_{opt}$ and T_j represent the inverted

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.



Fig. 6. Thermal infrared hyperspectral data cube acquired by ATHIS. (a) Field measurement targets for ground emissivity: quarry area and PTFE target. (b) Urban area. The data cube contains 320 spatial samples and 16 249 lines with a total of 110 spectral bands. The false-color composite image is synthesized based on three thermal infrared bands, with the red channel at 10.644 μm, green channel at 9.485 μm, and blue channel at 8.906 μm. This data cube was acquired at 13:14 China Standard Time (UTC+ 8) on May 24, 2019.

TABLE II
SPECIFICATIONS OF THE ATHIS INSTRUMENT

Parameter	Specification
Platform Type	Airborne Platform
Imaging Mode	Push-broom Hyperspectral Imaging
Spectral Range	8.0–12.5 μm
Number of Spectral Bands	155
Spectral Sampling Interval	38 nm
Sensitivity	0.17 K

LST and true LST of the samples in the dataset under evaluation, respectively. $i \in [1, N]$, where N is the number of spectral bands in the LSE spectrum. $\hat{\varepsilon}_{\text{opt}}(\lambda_i)$ and $\varepsilon_j(\lambda_i)$ represent the LSE and true LSE of sample j in band i, respectively.

B. Measured Data

To further evaluate the performance and practical applicability of the proposed OWTES algorithm under real observation conditions, thermal infrared hyperspectral data acquired by ATHIS were used for validation. The main specifications of the ATHIS instrument are summarized in Table II. The data cube was collected during an airborne remote sensing experiment conducted on May 24, 2019, over Dongyang City, Zhejiang Province, China (120°18′ 11″E, 29°09′ 20″N) [24]. The flight altitude was approximately 1000 m above ground level, corresponding to a ground spatial resolution of about 2 m. A synchronous ground measurement experiment was conducted using a polytetrafluoroethylene (PTFE) film as a reference target. During the experiment, weather conditions were clear and stable, with an air temperature of approximately 26.0 °C. The study area is located in a subtropical climate zone and features a complex landscape, including mountainous terrain and diverse land-cover types such as buildings, roads, lakes, rivers, abandoned quarries, grasslands, and farmlands, providing an ideal test site for algorithm validation (Fig. 6). The acquired thermal infrared hyperspectral data cube consists of 320 spatial samples $\times$16 249 lines $\times$110 spectral bands. Prior to TES retrieval, atmospheric correction was first performed on the original thermal infrared hyperspectral data to convert the sensor-received radiance into ground-leaving radiance. Following the atmospheric correction method described in [24], atmospheric transmittance, path radiance, and atmospheric downwelling radiance were estimated using a radiative transfer model driven by synchronous atmospheric parameters

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.

TABLE III
VARIATION OF LST RMSE WITH NEDT LEVEL BASED ON SIMULATED DATA FOR DIFFERENT TES ALGORITHMS

NEDT(K)	ARTEMISS(K)	WTTES(K)	RDSS(K)	OWTES(K)
0.00	0.09	0.94	0.14	0.14
0.05	0.22	0.95	0.18	0.20
0.10	0.39	0.98	0.25	0.29
0.20	0.74	1.05	0.38	0.45
0.30	1.13	1.13	0.59	0.60
0.40	1.52	1.22	0.76	0.75
0.50	1.93	1.32	0.94	0.88
0.60	2.32	1.42	1.13	1.06
0.70	2.72	1.52	1.33	1.22
0.80	3.21	1.63	1.54	1.38
0.90	3.69	1.74	1.80	1.55
1.00	4.26	1.84	2.07	1.69

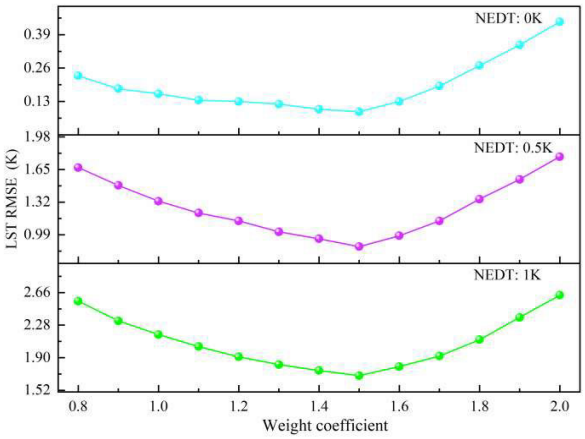


Fig. 7. Comparison of temperature errors under different weighting coefficients.

acquired during the experiment. The ground-leaving radiance was subsequently calculated according to the thermal infrared radiative transfer equation and used as the input for subsequent temperature and emissivity retrieval.

IV. RESULTS

A. Ozone Band Weight Analysis

If all spectral bands are assigned the same weight, the atmosphere–surface interaction information contained in the ozone absorption band cannot be fully exploited in multiband joint inversion. To address this issue, this study gradually increases the weight assigned to the ozone absorption region, thereby explicitly enhancing the contribution of ozone-related errors to the overall cost function and better capturing the unique radiative characteristics of this band during the iterative optimization process. To determine an appropriate weighting factor, temperature retrieval errors were evaluated under noise levels of 0, 0.5, and 1 K. As shown in Fig. 7, an ozone-band weight of 1.5 consistently yields the minimum temperature retrieval error across all noise conditions. Accordingly, a weight coefficient of 1.5 is adopted in this study.

B. Evaluation With Simulated Data

We evaluated the retrieval accuracy of four TES algorithms—OWTES, RDSS, ARTEMISS, and WTTES. The relationships between LST/LSE retrieval performance and noise levels are illustrated in Fig. 8, and the corresponding quantitative errors are summarized in Tables III and IV.

Overall, retrieval errors for all algorithms increase with increasing noise levels. A closer examination of the four TES algorithms indicates that OWTES achieves the lowest LST retrieval errors in most cases, although its advantage is not pronounced under noise-free conditions. At very low noise levels, its LST accuracy is slightly lower than that of RDSS. However, as noise increases, the performance gap between the two methods gradually widens. At a noise level of 1 K, OWTES reduces the LST error by approximately 0.38 K relative to RDSS, corresponding to an improvement of about 18.4%. The WTTES algorithm exhibits relatively stable error performance overall and ranks second only to OWTES at a noise level of 1 K. However, its performance degrades markedly under noise-free conditions, where LST errors approach 1 K, substantially higher than those of the other three algorithms. This behavior can be attributed to the reliance of WTTES on wavelet decomposition to separate high-frequency interference from the low-frequency emissivity component. Under high-noise conditions, strong high-frequency noise serves as an effective "anchor," facilitating the identification and removal of interference. Moreover, when surface emissivity is high, the contribution of atmospheric downwelling radiance to the measured signal is relatively small, which further stabilizes the separation process. In contrast, under low-noise conditions, this "anchor" effect diminishes, and the algorithm's limited sensitivity to atmospheric downwelling radiance makes it difficult to distinguish subtle high-frequency emissivity features from residual interference. Consequently,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.

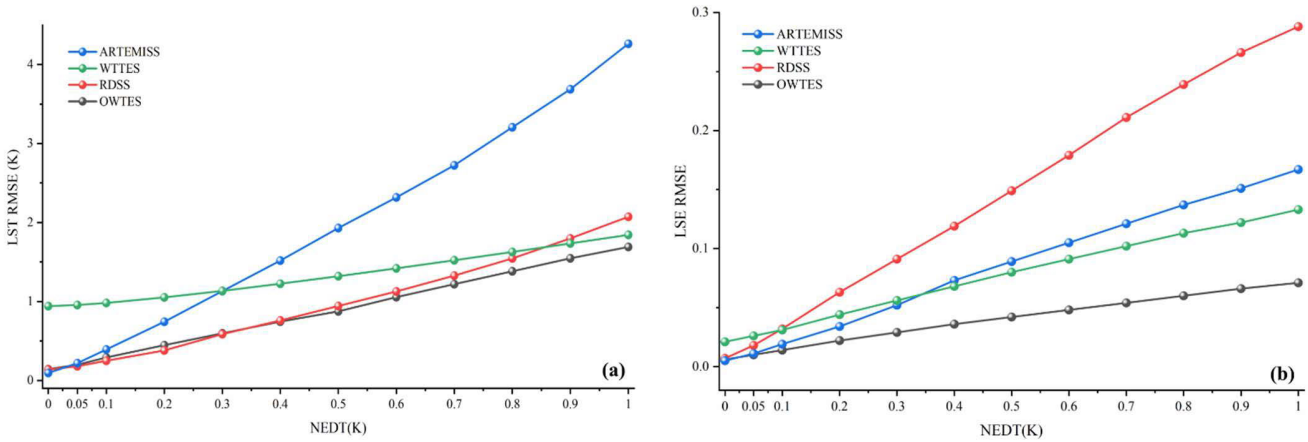


Fig. 8. Variation of LST and LSE retrieval errors with NEDT for different TES algorithms. (a) LST. (b) LSE. The results were validated on simulated data.

TABLE IV
VARIATION OF LSE RMSE WITH NEDT LEVEL BASED ON SIMULATED DATA FOR DIFFERENT TES ALGORITHMS

NEDT(K)	ARTEMISS	WTTES	RDSS	OWTES
0.00	0.005	0.021	0.007	0.006
0.05	0.011	0.026	0.018	0.010
0.10	0.019	0.031	0.032	0.014
0.20	0.034	0.044	0.063	0.022
0.30	0.052	0.056	0.091	0.029
0.40	0.069	0.068	0.119	0.036
0.50	0.082	0.080	0.149	0.042
0.60	0.095	0.091	0.179	0.048
0.70	0.116	0.102	0.211	0.054
0.80	0.131	0.113	0.239	0.060
0.90	0.146	0.122	0.266	0.066
1.00	0.159	0.133	0.288	0.071

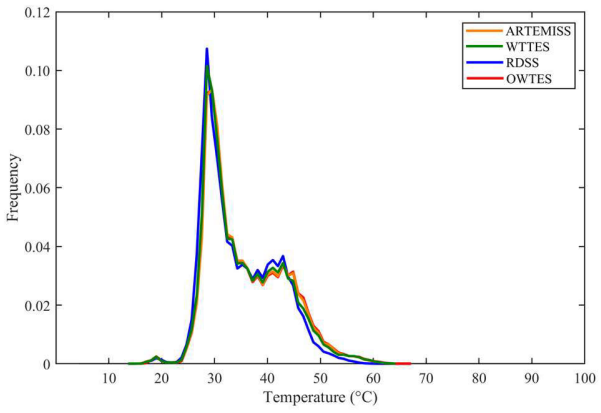


Fig. 9. Temperature histograms of different TES algorithms.

wavelet misclassification may occur, leading to distortion of the LSE spectrum and increased retrieval errors.

The ARTEMISS algorithm exhibits relatively high temperature retrieval accuracy under noise-free conditions, with an LST error of only 0.09 K, which is the lowest among all algorithms. However, as the noise level increases, its temperature error rises rapidly, indicating strong sensitivity to noise. When the NEDT increases from 0 to 1 K, the LST RMSE of ARTEMISS increases sharply from 0.09 to 4.26 K. Except for ISSTES, ARTEMISS exhibits the second fastest error growth among all algorithms, with its error increasing much more rapidly than those of RDSS, WTTES, and OWTES. This behavior indicates that ARTEMISS has relatively limited robustness to noise disturbances.

In terms of LSE retrieval accuracy, OWTES consistently achieves substantially lower LSE RMSE than the other algorithms. Moreover, the performance gap between OWTES and the remaining methods gradually widens as the noise level increases, indicating that OWTES is able to maintain high emissivity retrieval accuracy even under severe noise conditions. At an NEDT of 1 K, the LSE RMSE of OWTES is 0.062 lower than that of WTTES, which exhibits the best performance among the remaining algorithms, corresponding to an accuracy improvement of approximately 46.6%.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.

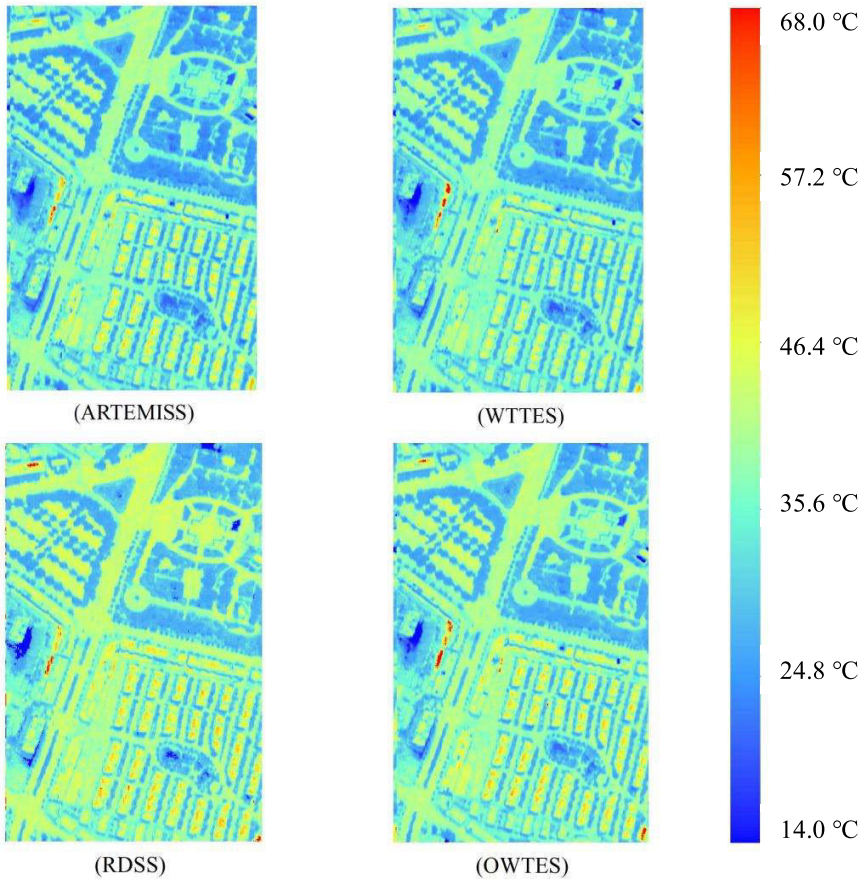


Fig. 10. Spatial distributions of temperature over selected areas retrieved from the ATHIS data cube.

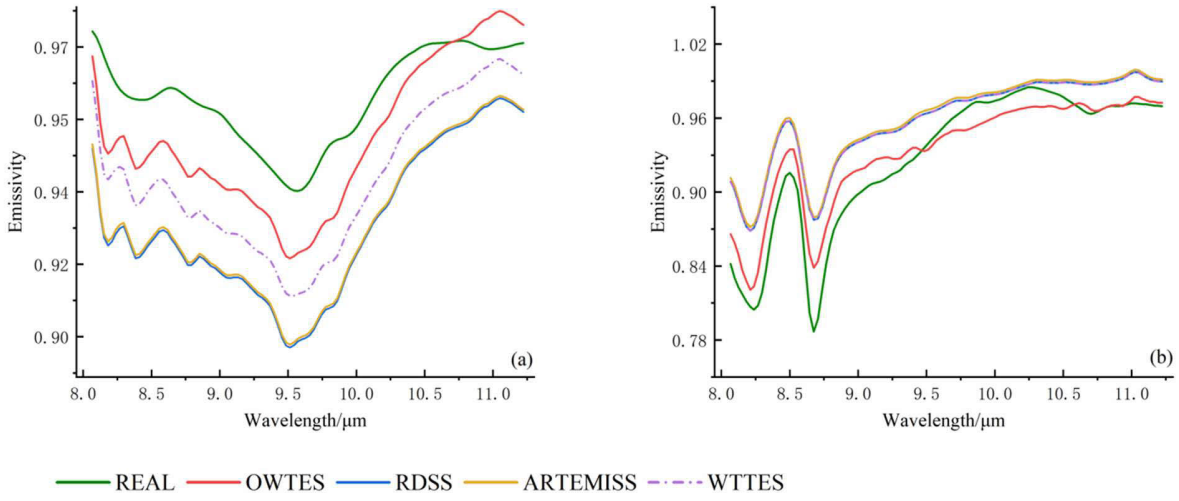


Fig. 11. Emissivity retrieval results of different TES algorithms for the target sites. (a) Quarry. (b) PTFE.

In contrast, RDSS shows relatively strong sensitivity to noise. Its LSE RMSE increases rapidly with increasing noise level, rising from 0.007 at NEDT = 0 K to 0.288 at NEDT = 1 K. Except for ISSTES, RDSS exhibits the second fastest error growth among all algorithms, indicating relatively limited stability under high-noise conditions. This behavior is likely related to the synchronized smoothing and resolution-degradation strategy employed by RDSS. Under strong noise conditions, random fluctuations can significantly distort the spectral structure, while excessive smoothing may suppress genuine emissivity details, ultimately leading to the accumulation of retrieval errors.

ARTEMISS and WTTES exhibit relatively similar overall performance in LSE retrieval. ARTEMISS achieves slightly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.

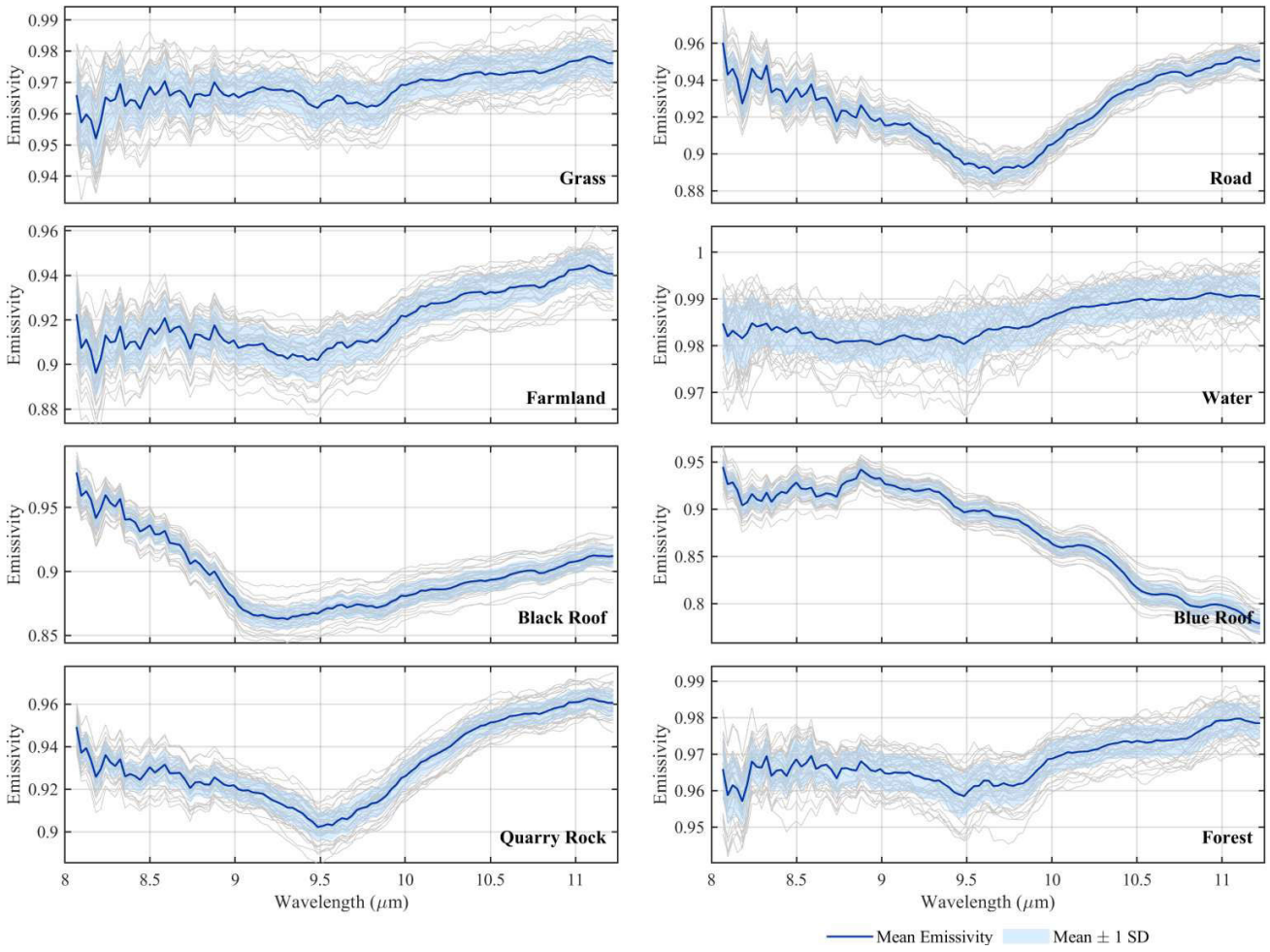


Fig. 12. Mean emissivity spectra and standard deviation ranges of different land-cover types retrieved by OWTES.

higher accuracy under low-noise conditions; for example, at NEDT = 0 K, its LSE RMSE is only 0.005, which is the lowest among all algorithms except OWTES. However, its error increases more rapidly as the noise level rises, reaching 0.159 at NEDT = 1 K. In comparison, although WTTES exhibits slightly larger errors than ARTEMISS under low-noise conditions, its error growth remains more gradual and becomes comparatively more stable under high-noise conditions. For instance, at NEDT = 1 K, the LSE RMSE of WTTES is 0.133, lower than the 0.159 obtained by ARTEMISS. This suggests that the wavelet-based noise suppression mechanism employed by WTTES is able to preserve the stability of emissivity spectra to some extent under strong noise conditions, thereby slowing the growth of retrieval errors.

Overall, OWTES provides consistent improvements in both LST and LSE retrieval accuracy, particularly under high-noise conditions, where it demonstrates significantly enhanced inversion performance.

C. Evaluation With Measured Hyperspectral Data Cube

1) Comparison of LST Retrieval Results: Fig. 9 presents the histograms of LST retrieved by different TES algorithms, and Fig. 10 shows the regional temperature image retrieved from the ATHIS data cube. Table V summarizes the statistical parameters of the temperature histograms. The dominant land-cover type in the data cube is forest; therefore, the main temperature peak of approximately 28 °C in Fig. 9 can be regarded as primarily representing the forest canopy temperature. This temperature differs from the simultaneously recorded air temperature (26 °C) by approximately 2 °C, which is generally consistent with previous studies on the relationship between vegetation canopy temperature and air temperature [25], indicating that the retrieved temperature results in this study are reasonably reliable. Overall, the temperature distributions retrieved by different TES algorithms are generally similar, with the main differences mainly appearing in the local temperature details of a certain surface type.

2) Comparison of LSE Retrieval Results: As shown in Fig. 11, the emissivity spectra retrieved by different algorithms for the quarry site and the PTFE target are presented, respectively. Table VI summarizes the statistical results of emissivity retrieval errors for the corresponding targets.

The in situ emissivity spectra were measured using a 102F Fourier transform infrared spectrometer. The green curve in Fig. 11 represents the measured emissivity and serves as a reference for evaluating the retrieval results of different TES

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.

TABLE V
STATISTICAL PARAMETERS OF TEMPERATURE HISTOGRAMS OF THE TES ALGORITHMS

Algorithm	Minimum(℃)	Maximum(℃)	Mean(℃)	Standard deviation(℃)
ARTEMISS	14.6	63.9	32.29	5.28
WTTES	14.1	64.6	33.51	4.13
RDSS	15.1	65.2	34.16	5.68
OWTES	14.7	67.2	34.02	4.36

TABLE VI
ACCURACY OF EMISSIVITY RETRIEVAL AT THE QUARRY SITE (PTFE TARGET AND QUARRY ROCK)

Algorithm	Quarry		PTFE	
	RMSE	Bias	RMSE	Bias
ARTEMISS	0.033	0.035	0.040	0.043
WTTES	0.023	0.022	0.038	0.037
RDSS	0.019	0.020	0.021	0.022
OWTES	0.015	0.013	0.018	0.015

algorithms. Although the emissivity spectra retrieved by the OWTES algorithm exhibit spectral shapes generally consistent with those obtained by the other three algorithms, OWTES yields the smallest retrieval errors among them.

Further error analysis shows that the emissivity retrieval errors for the two targets are 0.015 and 0.018, respectively, with corresponding mean biases of 0.013 and 0.014. Compared with the RDSS, ARTEMISS, and WTTES algorithms, the average retrieval errors of OWTES are reduced by approximately 0.005, 0.023, and 0.016, corresponding to relative improvements of about 25%, 61%, and 52%, respectively. Meanwhile, the retrieval biases are reduced by 0.007, 0.025, and 0.016, representing relative reductions of 33%, 64%, and 53%, respectively. These results indicate that OWTES achieves higher emissivity retrieval accuracy under observed data conditions, confirming its reliability in practical applications.

Fig. 12 presents the retrieved emissivity spectra from the measured data cube and their statistical distributions for different land-cover types. The results show that different surface types exhibit distinct spectral patterns within the 8–11.2 μm range, demonstrating that OWTES is able to effectively preserve the major spectral features of each land-cover type with good stability and consistency.

V. CONCLUSION

Based on the core design concept of enhancing the weighting of ozone-sensitive spectral bands, the OWTES algorithm is proposed in this study. Compared with existing TES methods, OWTES exhibits stronger stability under high-noise conditions. By strengthening the error constraints within ozone-sensitive spectral bands, OWTES effectively suppresses the interference of noise in the matching process between simulated and observed radiance, thereby avoiding parameter distortions caused by the combined effects of noise and ozone absorption in conventional algorithms. As a result, the retrieval accuracy of both LST and LSE is significantly improved.

Experimental results demonstrate that, under low-temperature conditions, OWTES improves the LST retrieval accuracy by approximately 0.75 K compared with RDSS. In terms of LSE retrieval, OWTES improves the accuracy by about 0.104 relative to WTTES, corresponding to a relative improvement of approximately 58%. In the validation using observed data, OWTES also achieved higher emissivity retrieval accuracy. Compared with ARTEMISS, WTTES, and RDSS, OWTES produced lower RMSE and bias values for both the quarry site and the PTFE target. In addition, the retrieved emissivity spectra showed good agreement with the measured spectra and were able to preserve the spectral characteristics of different surface targets more accurately. These results indicate that OWTES maintains good stability and reliability under practical observation conditions.

Although OWTES shows favorable retrieval performance under the experimental settings considered in this study, its dependence on spectral resolution and atmospheric correction accuracy still requires further evaluation. Future work will therefore focus on two aspects: 1) assessing the applicability of OWTES under different spectral resolution configurations to evaluate its generalization potential across various thermal infrared hyperspectral sensors and 2) further investigating the influence of atmospheric correction uncertainties on retrieval performance, with the aim of enhancing the robustness of OWTES in complex scenes.

REFERENCES

[1] Z.-L. Li et al., "Satellite remote sensing of global land surface temperature: Definition, methods, products, and applications," *Rev. Geophysics*, vol. 61, no. 1, pp. 1–77, Mar. 2023, doi: 10.1029/2022rg000777.

[2] Z.-L. Li et al., "Satellite-derived land surface temperature: Current status and perspectives," *Remote Sens. Environ.*, vol. 131, pp. 14–37, Apr. 2013, doi: 10.1016/j.rse.2012.12.008.

[3] H. Wu, X. Li, Z. Li, S. Duan, and Y. Qian, "Hyperspectral thermal infrared remote sensing: Current status and perspectives," *Nat. Remote Sens. Bull.*, vol. 25, no. 8, pp. 1567–1590, 2021, doi: 10.11834/jrs.20211306.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.

[4] G. Guo, B. Liu, and C. Liu, "Thermal infrared spectral characteristics of bunker fuel oil to determine oil-film thickness and API," *J. Mar. Sci. Eng.*, vol. 8, no. 2, p. 135, Feb. 2020, doi: 10.3390/jmse8020135.

[5] Z.-L. Li et al., "Land surface emissivity retrieval from satellite data," *Int. J. Remote Sens.*, vol. 34, nos. 9–10, pp. 3084–3127, Oct. 2012, doi: 10.1080/01431161.2012.716540.

[6] S. Meerdink et al., "Plant species' spectral emissivity and temperature using the hyperspectral thermal emission spectrometer (HyTES) sensor," *Remote Sens. Environ.*, vol. 224, pp. 421–435, Apr. 2019, doi: 10.1016/j.rse.2019.02.009.

[7] J. A. Sobrino and N. Raissouni, "Toward remote sensing methods for land cover dynamic monitoring: Application to Morocco," *Int. J. Remote Sens.*, vol. 21, no. 2, pp. 353–366, Nov. 2010.

[8] C. L. Li et al., "Development of infrared hyperspectral remote sensing imaging and application of gas detection (invited)," *Infr. Laser Eng.*, vol. 51, no. 7, pp. 33–45, Jul. 2022, doi: 10.3788/IRLA20210866.

[9] J. A. Hackwell et al., "LWIR/MWIR imaging hyperspectral sensor for airborne and ground-based remote sensing," in *Proc. SPIE*, Denver, CO, USA, 1996, pp. 102–107, doi: 10.1117/12.258057.

[10] G. C. Hulley et al., "High spatial resolution imaging of methane and other trace gases with the airborne hyperspectral thermal emission spectrometer (HyTES)," *Atmos. Meas. Techn.*, vol. 9, no. 5, pp. 2393–2408, Jun. 2016, doi: 10.5194/amt-9-2393-2016.

[11] Z. Aslett, J. V. Taranik, and D. N. Riley, "Mapping rock-forming minerals at daylight pass, Death Valley national park, California, using SEBASS thermal-infrared hyperspectral image data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Boston, MA, California, Jul. 2008, pp. 366–369.

[12] J. Cui et al., "Temperature and emissivity separation and mineral mapping based on airborne TASI hyperspectral thermal infrared data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 40, pp. 19–28, Aug. 2015, doi: 10.1016/j.jag.2015.03.014.

[13] R. G. Vaughan, W. M. Calvin, and J. V. Taranik, "SEBASS hyperspectral thermal infrared data: Surface emissivity measurement and mineral mapping," *Remote Sens. Environ.*, vol. 85, no. 1, pp. 48–63, Apr. 2003, doi: 10.1016/s0034-4257(02)00186-4.

[14] A. Palombo et al., "Soil moisture retrieval by integrating TASI-600 airborne thermal data, WorldView 2 satellite data and field measurements: Petacciato case study," *Sensors*, vol. 19, no. 7, p. 1515, Mar. 2019, doi: 10.3390/s19071515.

[15] S. Pascucci, S. Pignatti, R. Casa, R. Darvishzadeh, and W. Huang, "Special issue 'hyperspectral remote sensing of agriculture and vegetation,"' *Remote Sens.*, vol. 12, no. 21, p. 3665, Nov. 2020, doi: 10.3390/rs12213665.

[16] W. R. Johnson, G. Hulley, and S. J. Hook, "Remote gas plume sensing and imaging with NASA's hyperspectral thermal emission spectrometer (HyTES)," in *Proc. SPIE*, vol. 9101, Baltimore, MD, USA, 2014, pp. 91010V-1–91010V-7, doi: 10.1117/12.2049005.

[17] A. Mukundan, C. C. Huang, T. C. Men, F. C. Lin, and H. C. Wang, "Air pollution detection using a novel snap-shot hyperspectral imaging technique," *Sensors*, vol. 22, no. 16, p. 6231, Aug. 2022, Art. no. 6231, doi: 10.3390/s22166231.

[18] C. Borel, "Iterative retrieval of surface emissivity and temperature for a hyperspectral sensor," in *Proc. 1st JPL Workshop Remote Sens. Land Surf. Emissivity*, Pasadena, CA, USA, Nov. 1997, pp. 6–8.

[19] C. Borel, "Error analysis for a temperature and emissivity retrieval algorithm for hyperspectral imaging data," *Int. J. Remote Sens.*, vol. 29, nos. 17–18, pp. 5029–5045, Sep. 2010, doi: 10.1080/01431160802036540.

[20] C. Borel, D. Rosario, and J. Romano, "Data processing and temperature-emissivity separation for tower-based imaging Fourier transform spectrometer data," *Int. J. Remote Sens.*, vol. 36, nos. 19–20, pp. 4779–4792, Jun. 2015, doi: 10.1080/01431161.2015.1047996.

[21] H. Shao, C. Liu, F. Xie, C. Li, and J. Wang, "Noise-sensitivity analysis and improvement of automatic retrieval of temperature and emissivity using spectral smoothness," *Remote Sens.*, vol. 12, no. 14, p. 2295, Jul. 2020, doi: 10.3390/rs12142295.

[22] Y.-Z. Zhang, H. Wu, X.-G. Jiang, Y.-Z. Jiang, Z.-X. Liu, and F. Nerry, "Land Surface Temperature and emissivity retrieval from field-measured hyperspectral thermal infrared data using wavelet transform," *Remote Sens.*, vol. 9, no. 5, p. 454, May 2017, doi: 10.3390/rs9050454.

[23] H. Shao, C. Liu, C. Li, J. Wang, and F. Xie, "Temperature and emissivity inversion accuracy of spectral parameter changes and noise of hyperspectral thermal infrared imaging spectrometers," *Sensors*, vol. 20, no. 7, p. 2109, Apr. 2020, doi: 10.3390/s20072109.

[24] C. Liu et al., "New airborne thermal-infrared hyperspectral imager system: Initial validation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4149–4164, 2020, doi: 10.1109/JSTARS.2020.3010092.

[25] N. Dong, I. C. Prentice, S. P. Harrison, Q. H. Song, and Y. P. Zhang, "Biophysical homoeostasis of leaf temperature: A neglected process for vegetation and land-surface modelling," *Global Ecology Biogeography*, vol. 26, no. 9, pp. 998–1007, Sep. 2017, doi: 10.1111/geb.12614.



Xia Zhang received the B.S. degree in GIS from Changchun Normal University, Changchun, China, in 2008, and the M.S. and Ph.D. degrees in cartography and geographic information systems from Northeast Normal University, Changchun, in 2010 and 2014, respectively.

From 2014 to 2020, she was a Lecturer, and since 2021, she has been an Associate Professor with the School of Land Science and Spatial Planning, Hebei GEO University, Shijiazhuang, Hebei, China. Her research interests include thermal infrared remote sensing, agricultural remote sensing, and quantitative remote sensing.



Ruohan Chen received the B.S. degree in geographic information systems from Hebei GEO University, Shijiazhuang, Hebei, China, in 2023, where she is currently pursuing the M.S. degree in resources and environment.

Her research interests include thermal infrared remote sensing.



Chunlai Li received the B.S. degree in optoelectronic engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2004, and the Ph.D. degree in electronic science and technology from Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai, China, in 2009.

He is a Researcher with the Shanghai Institute of Technical Physics, Chinese Academy of Sciences. His research interests include weak infrared spectrum signal processing, refrigeration infrared focal plane information acquisition, and infrared hyperspectral data processing.



Guofei Shang received the B.E. degree in aerial photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1985, and the M.E. and Ph.D. degrees in land resources management from the Renmin University of China, Beijing, China, in 1992 and 2002, respectively.

Since 1997, he has been a Professor and the Dean of the College of Land Resources and Urban and Rural Planning, Hebei GEO University, Shijiazhuang, China. His research interests include aerial photogrammetry and remote sensing, quantitative remote sensing, and natural resources management.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.



Zhe Li received the B.S. degree in land management from the Renmin University of China, Beijing, China, in June 1992.

From 2006 to 2002, she was a Senior Engineer, and since 2023, she has been a Professor of engineering. Her research interests include natural resources development and utilization, land grading and valuation, intensive land use evaluation, and related areas.



Yongli Chen received the B.S. degree in geodetic surveying from Wuhan University of Surveying and Mapping, Wuhan, China, in 1997, and the M.S. degree in surveying and mapping engineering from Wuhan University, Wuhan, in 2009.

He is currently a Senior Engineer at the Second Surveying and Mapping Institute of Hebei Province, Shijiazhuang, China. His research interests include satellite navigation and positioning, quasi-geoid, application of satellite remote sensing technology, and GIS fields.



Haoran Xiong received the B.S. degree in surveying and mapping engineering from Hebei GEO University, Shijiazhuang, China, in 2024, where he is currently pursuing the M.S. degree in resources and environment.

His research interests include quantitative remote sensing.



Suya Zhao received the B.S. degree in geographic information systems from Hebei GEO University, Shijiazhuang, Hebei, China, in 2023, where she is currently pursuing the M.S. degree in resources and environment.

Her research interests include thermal infrared remote sensing.



Chengyu Liu (Member, IEEE) received the B.S. degree in GIS from Northeast Normal University, Changchun, China, in 2008, and the Ph.D. degree in natural resources from Beijing Normal University, Beijing, China, in 2013.

He is currently a Senior Engineer with Key Laboratory of Spatial Active Opto-Electronic Techniques, Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai, China. His current interests include calibration and data processing of infrared hyperspectral remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:59:19 UTC from IEEE Xplore. Restrictions apply.